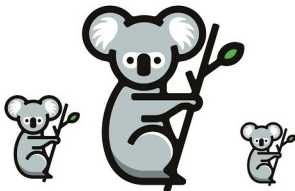


OpenStack Kolla: Building the Community Around Containerised OpenStack Deployment

Kolla provides production-ready containers and deployment tools for operating OpenStack clouds that are scalable, fast, reliable and upgradable — using community best practices.



I still remember the kind of applause I got from my colleagues when I got my first OpenStack deployment working. Most of the things then worked around DevStack, the development environment of OpenStack, and vendors were still trying to find the right mix of DevOps tools to deploy OpenStack. The evolution process involved working with tools like Chef, Puppet and Ansible, with vendors creating a deployment model around them. Then came Docker, creating a buzz with a lost Linux feature called containers. A group of community members came together to see if they could containerise the OpenStack services and deploy them. The project started with the sole mission of containerising OpenStack into microservices and providing additional tooling to simplify management.

The entry barrier for Kolla

The arrival of Kolla was not particularly welcomed by the community, which was busy with already proven technologies to automate the OpenStack deployment process. At the same time, the Kolla team had its own set of challenges to cope with. Docker was itself very new at this stage, with very limited automation frameworks to use. The project went through a set of repeated events, while rediscovering the architectural changes and tools to be used. The initial implementation focused on creating the individual Docker images for core services for Centos and Ubuntu operating system flavours and deploying them with Kubernetes, which was just picking up. The configuration management part was still a half-solved puzzle, and networking was far below the requirements of both OpenStack deployment as well as its users. The deployment part was then rewritten in *docker-compose*, which at that

time was the most popular deployment framework for multiple containers to add one more version of project rewrites.

Deploying containerised OpenStack with Kolla and Ansible

With the time spent fiddling around with the OpenStack Docker images, the Kolla team decided to reinvent the architecture again, and look for alternatives that could help deploy the containerised services with better control. This time, the team turned towards Ansible—at that time one of the latest configuration management tools on the horizon; it was still awaiting wider community adoption, so there were few preconceived notions about containers. The Kolla team used the same container images and a few known bits of the Ansible framework to get the first successful deployment. The configuration management part was well managed with Ansible. During the process, the team made a lot of changes in the Ansible Docker module, which was then contributed back to the Ansible project.

Making baby steps towards wider community adoption

The initial deployment of Kolla with Ansible was an achievement that attracted a lot of attention towards Kolla as a deployment framework for OpenStack, resulting in a number of contributors and innovative ideas from users who did different PoCs (proofs-of-concept). The team's diversity increased and it was accepted as a part of the Big Tent governance model of the OpenStack Foundation. The Docker images were central to the adoption of Kolla and the image creation got a major revamp. Jinja based image templates took the place of individual Docker files and a configuration based build framework to build specific images. The same framework was again extended to include